

Shashwat Tiwari

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Education

Northeastern University, Boston
MS Computer Science **GPA: 4.0**

September 2024 – May 2026

Shiv Nadar University, India
BS Computer Science **GPA: 3.8**

Aug 2020 – May 2024

Relevant Coursework

Algorithms, Object Oriented Programming, Database Systems, Data Visualization, Cloud Computing(AWS ACA & ACF), Machine Learning, Web Development, Artificial Intelligence, Software Engineering(SDLC)

Skills

Programming & Statistical Languages: Python, R, SQL, Java, C/C++ , Git

BI & Visualization: Tableau, Power BI, Excel **ML & Analytics:** PyTorch, TensorFlow, Scikit-learn, XGBoost, SageMaker, Hugging Face Transformers, LangChain

Cloud & DevOps : AWS, Azure, Docker, CI/CD Pipelines, Agile/Scrum

Experience

Data Analyst Intern, Bain & Company

January 2024 – July 2024

- Designed and deployed end-to-end ML pipelines for M&A deal screening, reducing manual analyst review time by 40% through automated target segmentation
- Built a time-series forecasting model (ARIMA + XGBoost ensemble) to predict M&A activity signals across industries, improving prospect prioritization accuracy by 18% over rule-based heuristics
- Engineered automated SQL + Python (Pandas, SQLAlchemy) ingestion pipeline with schema validation, maintaining data accuracy across a 1,000+ prospect corpus
- Developed NLP classification models (TF-IDF + Logistic Regression) for insight extraction from M&A deal articles cutting manual labeling overhead by 35%

Software Engineer Intern, Aftershoot Inc.

May 2023 – July 2023

- Architected a high-throughput data ingestion pipeline in Python (Pandas, NumPy) to ingest and normalize subscription and usage telemetry from the Stripe API, enabling downstream analytics on customer retention and churn
- Refactored core ETL modules from Python to Rust, achieving a 3x reduction in processing time and cutting peak memory usage by 45%
- Designed data schemas and processing logic to support product and data science teams, directly informing feature prioritization decisions based on usage pattern analysis

Projects

RegTranslate - SaaS Compliance Tool

[Python, FastAPI, React, Llama 3, LangChain, Docker, Jira API]

- Engineered a RAG-based compliance platform that ingests regulatory PDFs (HIPAA, GDPR) and auto-generates structured developer tasks (Jira tickets) using Llama 3 and LangChain, streamlining compliance workflows for engineering teams
- Built a semantic search pipeline using ChromaDB vector store and HuggingFace embeddings, optimizing context retrieval latency and achieving high-accuracy task generation from large, ambiguous legal corpora
- Developed a React dashboard for real-time document ingestion and task tracking, backed by a FastAPI async backend with chunked document processing

Multi-Agent Research Assistant

[Python, LangGraph, ChromaDB, MCP]

- Built a multi-agent research pipeline using LangGraph to orchestrate stateful agent graphs with specialized roles (query planning, document retrieval, answer synthesis) for automated research workflows
- Integrated ChromaDB with sentence-transformer embeddings for semantic memory across research sessions, and exposed the full pipeline as an MCP server for invocation from MCP-compatible AI clients

StaffSync – Employee Management & Analytics Platform

[Python, Django, HTML, MySQL]

- Built a role-based employee management platform using **Python/Django + MySQL**, with role based dashboards and access
- Designed normalized relational schema with optimized queries; implemented **stored procedures, triggers, and cascading constraints** for automated payroll, leave, and audit logging
- Integrated analytics dashboards (Matplotlib) to track workforce productivity, trends, and retrieve data-driven insights